In-Person Worksheet: Unit 2 Summative

Polynomials and Rational Functions

Name: _____ Date: _____

Easy problem first. If you get it, try the hard one.

1. End behavior in limit form

1E. $f(x) = -3x^5 + 2x^2 - 7$. Write the end behavior using limits: $\lim_{x\to\infty} f(x) =$? and $\lim_{x\to-\infty} f(x) =$?

1H. $g(x) = (2x-1)^3(x+4)^2(5-x)$

- (a) What are the degree and leading coefficient? *(Don't multiply it all out.)*
- (b) Write the end behavior using limits.
- (c) A different polynomial rises on the left and falls on the right. What do you know about its degree and leading coefficient?

2. Polynomial factoring

2E. Factor completely by grouping: $x^3 - 2x^2 - 9x + 18$

2H.

- (a) Factor completely: $x^4 - 5x^2 - 36$ *(Hint: treat it like a quadratic in x^2.)*
- (b) $x = 2$ is a zero of $p(x) = 2x^3 + 3x^2 - 11x - 6$. Use synthetic division to factor p completely.

3. Roots of polynomials (including imaginary)

3E. Solve $x^2 - 4x + 13 = 0$.

3H. A polynomial with real coefficients has zeros $x = 1$ and $x = 2 - i$.

- (a) What other zero must it have, and why?
- (b) Write the polynomial of least degree with these zeros and leading coefficient 1, in standard form.
- (c) How many x-intercepts does its graph have?
- (d) List **all** zeros of $x^4 - 5x^2 - 36$ from 2H(a).

4. Multiplicity

4E. $f(x) = (x + 2)^2(x - 1)^3(x - 4)$

- (a) List each zero with its multiplicity. Does the graph cross or bounce at each one?
- (b) Degree and end behavior (in limit form)?

4H. Write the polynomial of least degree whose graph **bounces** at $x = -1$, **crosses** at $x = 2$, and has y-intercept $(0, -8)$.

- (a) Find it in factored form. (b) End behavior in limit form. (c) Sketch it.

5. Rational functions: characteristics

5E. $r(x) = \dfrac{x-3}{x+2}$. Fill in the table.

Vertical asymptote:	
Domain:	
Horizontal asymptote:	
Range:	
x-int:	
y-int:	

5H. $r(x) = \dfrac{2x^2-8}{x^2-x-6}$. Fill in the table. *(Factor first. Is there a hole?)*

Hole:	
Vertical asymptote:	
Domain:	
Horizontal asymptote:	
Range:	
x-int:	
y-int:	

6. Rational functions: graphs

6E. Sketch $r(x) = \dfrac{2x-1}{x-1}$. Label the asymptotes and intercepts.

6H. $r(x) = \dfrac{x+1}{x^2-4}$

- (a) Find the vertical asymptotes, horizontal asymptote, and intercepts.
- (b) Make a sign chart: is r positive or negative on each interval?
- (c) Use limits to describe what happens on each side of each vertical asymptote.
- (d) Sketch it. Does the graph cross its horizontal asymptote?

7. Rational functions: end behavior

7E. Find each limit.

- (a) $\lim_{x\to\infty} \dfrac{3x^2+1}{x^2-5}$ (b) $\lim_{x\to-\infty} \dfrac{4x}{x^3+1}$ (c) $\lim_{x\to\infty} \dfrac{x^3}{2x^2+7}$

7H. $r(x) = \dfrac{x^2+x-4}{x-2}$

- (a) Divide to find the slant asymptote.
- (b) Write the end behavior in limit form.
- (c) Find $\lim_{x\to 2^+} r(x)$ and $\lim_{x\to 2^-} r(x)$.

Answer key

1E. Degree 5 (odd), leading coefficient -3 (negative): $\lim_{x\to\infty} f(x) = -\infty$, $\lim_{x\to-\infty} f(x) = \infty$.

1H. (a) Multiply the leading terms only: $(2x)^3(x)^2(-x) = -8x^6$. Degree 6, leading coefficient -8. (b) Even degree, negative: $\lim_{x\to\infty} g(x) = -\infty$ and $\lim_{x\to-\infty} g(x) = -\infty$. (c) The ends go opposite ways, so the degree is **odd**; it falls on the right, so the leading coefficient is **negative**.

2E. $x^2(x-2) - 9(x-2) = (x-2)(x^2-9) = (x-2)(x-3)(x+3)$.

2H. (a) Let $u = x^2$: $u^2 - 5u - 36 = (u-9)(u+4)$, so $(x^2-9)(x^2+4) = (x-3)(x+3)(x^2+4)$. x^2+4 doesn't factor over the reals. (b) Synthetic division by 2 on $2, 3, -11, -6$ gives $2, 7, 3$ with remainder 0. So $p(x) = (x-2)(2x^2+7x+3) = (x-2)(2x+1)(x+3)$.

3E. $x = \dfrac{4 \pm \sqrt{16-52}}{2} = \dfrac{4 \pm 6i}{2} = 2 \pm 3i$.

3H. (a) $2+i$. Complex zeros of real polynomials come in conjugate pairs. (b) $(x-1)(x-(2-i))(x-(2+i)) = (x-1)(x^2-4x+5) = x^3-5x^2+9x-5$. (c) **One** ($x=1$); the imaginary zeros don't touch the x-axis. (d) From $(x-3)(x+3)(x^2+4)$: $x = 3, -3, 2i, -2i$.

4E. (a) $x=-2$, multiplicity 2: bounce. $x=1$, multiplicity 3: crosses, flattening out (S-shape). $x=4$, multiplicity 1: crosses. (b) Degree $2+3+1=6$, leading coefficient $+1$: $\lim_{x\to\pm\infty} f(x) = \infty$.

4H. (a) Bounce means even multiplicity, so $f(x) = a(x+1)^2(x-2)$. Then $f(0) = a(1)(-2) = -8$, so $a=4$: $f(x) = 4(x+1)^2(x-2)$. (b) Degree 3, positive: $\lim_{x\to\infty} f(x) = \infty$, $\lim_{x\to-\infty} f(x) = -\infty$. (c) Comes up from the bottom left, touches the axis at $x=-1$ and turns back down, passes through $(0,-8)$, crosses up through $x=2$.

5E. VA $x=-2$. Domain $x \neq -2$. HA $y=1$ (same degree: 1/1). Range $y \neq 1$. x-int $(3,0)$. y-int $(0,-\frac{3}{2})$.

5H. $r(x) = \dfrac{2(x-2)(x+2)}{(x-3)(x+2)} = \dfrac{2(x-2)}{x-3}$ for $x \neq -2$. Hole at $x=-2$: plug into the simplified form, $\dfrac{2(-4)}{-5} = \dfrac{8}{5}$, so the hole is $(-2, \frac{8}{5})$. VA $x=3$. Domain $x \neq -2, 3$. HA $y=2$ (same degree: 2/1). Range $y \neq 2$ and $y \neq \frac{8}{5}$ (the hole). x-int $(2,0)$; $x=-2$ is not one because it's the hole. y-int $(0, \frac{4}{3})$.

6E. VA $x=1$. HA $y=2$. x-int $(\frac{1}{2}, 0)$. y-int $(0,1)$. Two branches: lower-left (through the intercepts, heading down to $-\infty$ as $x \to 1^-$) and upper-right (coming down from ∞ as $x \to 1^+$, leveling to $y=2$).

6H. (a) VA $x=-2$ and $x=2$. HA $y=0$ (bottom degree is bigger). x-int $(-1,0)$. y-int $(0,-\frac{1}{4})$. (b) Test points: $(-\infty,-2)$ negative; $(-2,-1)$ positive; $(-1,2)$ negative; $(2,\infty)$ positive. (c) $\lim_{x\to-2^-} r = -\infty$, $\lim_{x\to-2^+} r = \infty$, $\lim_{x\to2^-} r = -\infty$, $\lim_{x\to2^+} r = \infty$. (d) Yes: it crosses $y=0$ at $x=-1$. A horizontal asymptote only describes the ends, so the graph can cross it in the middle.

7E. (a) Same degree: ratio of leading coefficients, 3. (b) Bottom degree bigger: 0. (c) Top degree bigger by one: ∞ (it follows a slant asymptote upward).

7H. (a) $x^2+x-4 = (x-2)(x+3)+2$, so $r(x) = x+3+\dfrac{2}{x-2}$. Slant asymptote $y=x+3$. (b) $\lim_{x\to\infty} r(x) = \infty$ and $\lim_{x\to-\infty} r(x) = -\infty$. (c) The numerator at $x=2$ is 2 (positive). From the right the denominator is a small positive number, so $\to \infty$. From the left it is a small negative number, so $\to -\infty$.